Good Manufacturing Practices (GMP) and Quality Awareness Training

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12/15//2024_AK

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Agenda

- Purpose of the Good Manufacturing Practices (GMP) Training Program

- Quality Awareness / Quality Management System (QMS)

  An Overview

- Good Manufacturing Practices (GMP's)

  Rules, guidelines and the importance to adhere to them.

- Good Documentation Practices (GDP's)
- Hygiene Training Program

Purpose of this Training Program

- Our **CUSTOMERS require** that we have regular GMP / Quality Awareness training
- Food and Drug Administration (FDA) regulations requires GMP trainings.
- GMP requires that **all personnel are well trained** and understand the concept of GMP.
- Our Quality Management System requires it.
- To provide an awareness of quality, our responsibility to it and our impact on it.

Quality Definition

- **Quality:** A subjective term for which each person or business sector has its own definition.

In technical terms, quality can have two meanings:

1. The characteristics of a product or service that bear on its ability to satisfy stated or implied needs
   - Not just product quality – every one of us provides a product or service to our downstream customer
2. A product or service free of deficiencies.

Fitness for use or conformance to specifications

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Babylon`s Quality Policy

- *Babylon is committed to a quality management system that improves customer satisfaction and promotes continuous improvement through employee involvement.*
- *We believe in a **Lean philosophy that places quality as our first priority** on each and every job.*
- *It is our basic operating philosophy to focus on preventing defects, making quality a way of life, and perpetuating an attitude of doing it right the first time.*

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Our Quality Management System (QMS) Structure and Overview

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The ISO and cGMP Standards:

- ISO 9001:2008
  - International Organization for Standardization
  - Internationally recognized
  - Customer driven
  - Defines requirements for a QMS
- ISO 22716:2007
  - Guidance for production, control, storage and shipment of cosmetic products
  - Internationally recognized
- cGMP – cosmetic GMP
  - Food and Drug Administration (FDA)
  - Nationally recognized
  - Compliance driven
  - Defines requirements for ensuring human consumable products are safe for use
  - Defines requirements for a QMS

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Current Good Manufacturing Practices

(cGMPs)

Good Manufacturing Practices (GMP) Training

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What are the "Good Manufacturing Practices (GMP)"?

It is a collection of strict guidelines and rules with one important purpose.

What is the purpose of GMP?
Main purpose is to protect the consumer.

GMP has a set rules to ensure that cosmetics are produced in a consistently good quality.

- GMP includes procedures to verify that the products are consistently high in quality, from batch to batch, for their intended use.
- The GMP rules include the monitoring of the quality during the entire production.
- The GMP rules apply to every process and procedure, from the starting point of the production to the finished and packed product.
- The rules regulate operational and personal hygiene.

It is all employee's responsibility to adhere to GMP all the time.

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Trust is essential. Consider the customers of pharmaceutical products...

- **People who are unwell.**
  - **sometimes very ill.**
- **Do not know if the product:**
  - **contains the right ingredient**
  - **contains the right dose**
  - **is contaminated**
  - **has degraded**

They must trust doctors, pharmacists and the pharmaceutical industry to do things right!
Trust is built by the quality of the product and services.

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Written Procedures

- Latest revision be indicated on the header.
- Approval Authority (Document owner, Quality and other stakeholders)
- Revision History must be indicated.
- No uncontrolled or obsolete documents posted or kept.

BABYLON	Document Name: Packaging Line Clearance Checklist Document Number: 7.3.1-1	Revision No: New Page: 1 of 3 Effective Date: 5/12/24
Reference Document Change Control for Approval Signatures of Process Owner and Approver		

Product:	SKU:
P.O Number:	P.O Quantity:
Batch Size (KG):	Date:

Production and Quality Assurance- Verify the following items:

Verification Description	Result	Production Initials	Quality Assurance Initials	Date
Correct Product: Confirm the SKU matches the work order. Does the Sku match what's in the work order?	Yes/No?			
Foreign Matter: Collect a lot of samples and visually inspect for any foreign matter that may be in the bulk product. Is there any foreign matter in the bulk product?	Yes/No?			
Correct Components: Are the Components in the transfer area the same as the Components on the work order?	Yes/No?			
Container (bottle,jar,tube etc..)	Item#:			
Cap/Top/Pump	Item#:			
Label	Item#:			

Proprietary Information of Our Babylon LLC – Not to be Disclosed or Reproduced without Prior Written Approval

Uncontrolled when Printed

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Follow Procedures

Not a suggestion – But a Requirement!

- Many customer non-conformances have been written for not following our procedures and instructions
- What happens when these documents are not followed?
- What are some excuses for not following them?

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Follow Procedures…

Reasons given for not following required documents:

- We don't know Why we should
- Unclear How
- Your way just won't work
- My way is better
- No Time
- Obstacles beyond my control
- Ineffective management support/control

If what we say we do isn't what we do – let's fix it

Ask questions and report your problems if you have difficulties to follow procedures.

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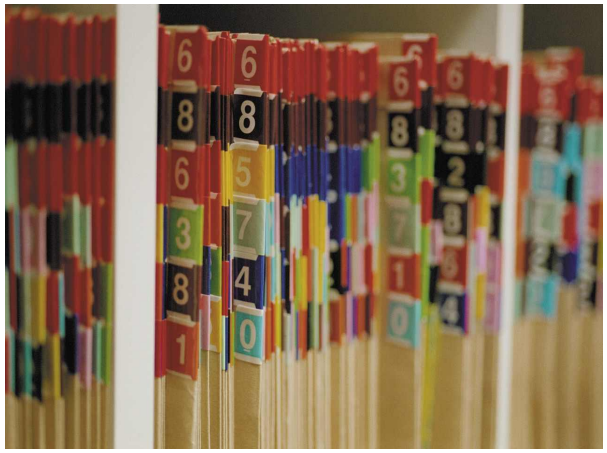
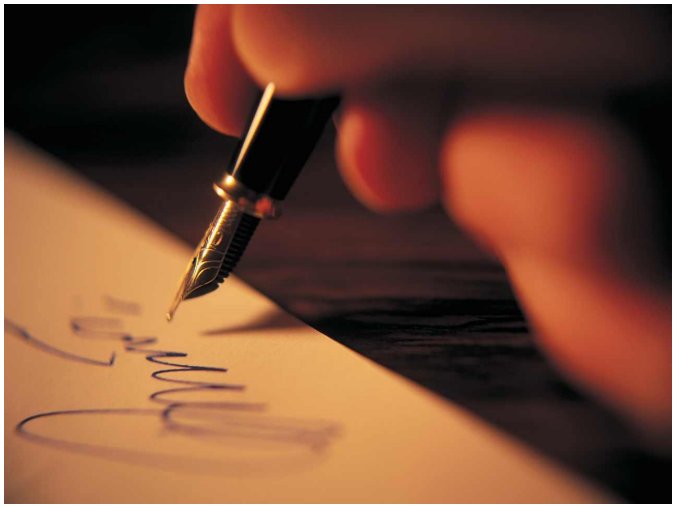
Document your work

Type of records:



- Batch Records
- Inspection Records
- Calibration Records
- Equipment Logs
- PM Records
- Complaint Records
- Internal Audit Reports
- Training Records



Document your work

Legible

Traceable

Clear

Make sure to use lots of TLC

Remember, If it's not documented, it wasn't Done

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Good Documentation Practices

All original hand-written entries shall be:

- Written legibly.
- Written using a blue or black permanent ballpoint pen.
- Written concurrently with work being performed.

All data shall be labeled with identifying information.

- Computer generated printouts should be labeled on at least the first page with project ID, test article, etc.
- If the printout does not include the date and time on each page, then the reporter's initials and date of signing are required on every page.

- All data shall be signed and dated concurrently with the work being performed.
- Backdating entries is not allowed

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Good Documentation Practices Examples:

To correct **your own** error, draw a single line through the entry then your initial and date.

Annotate the reason for the correction.
Assure that a clear and concise explanation is provided for the correction. For example: date error, spelling error, late entry.

- No overwrites
- No scribble outs
- No whiteout

- Write out all information
- No blank spaces
- Where not applicable write N/A
- Check N/A box if provided
- No ditto mark.

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Good Documentation Practices Examples:

Indicate the date correctly.

Dates should be formatted using either the first 3 letters of the month or the complete spelling instead of the numerical representation of the month. Because of country differences in recording dates, the **use of numbers can cause confusion** and possibly errors.

Example: The written date of 7/12/19 means July 12, 2019 in the USA and December 7, 2019 in most of Europe, and 2007 Dec 19th in Hungary and Japan.

Good example: Jul 12,2019 or July 12,2019

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Making new copies of forms

When making copies of a quality form:

- Check in Intelex to confirm the most recent revision is being used
- Ensure the form number, revision level and date appear on the copies
- Copies of original documents must be marked as "COPY"
- DO NOT MAKE COPIES OF A COPY!

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Control of Quality Records cont'd

- Storage of records
  - Store records to prevent damage, deterioration and loss
  - Quality Records Table – Responsibility - Supervisors/Managers
  - Describes how records are managed (retention time, method of destruction, storage location, responsibility, etc.)
  - Batch records at Babylon are kept for 5 years

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Facilities and Equipment

The facility must be of suitable size and layout for proper operation

- Production
- Storage
- Sanitary rooms, which consist of dress changing, washing, and toilet rooms,
- Cleaning
- Maintenance

Facility also includes:

- Lighting
- Ventilation
- Heating
- Cooling
- Containment

Production area must be separated from other areas and accessed only in the required attire and PPE.
The rooms should ensure a proper production process, without the possibility of contamination.

Interior surfaces of the rooms must be suitable for efficient cleaning.

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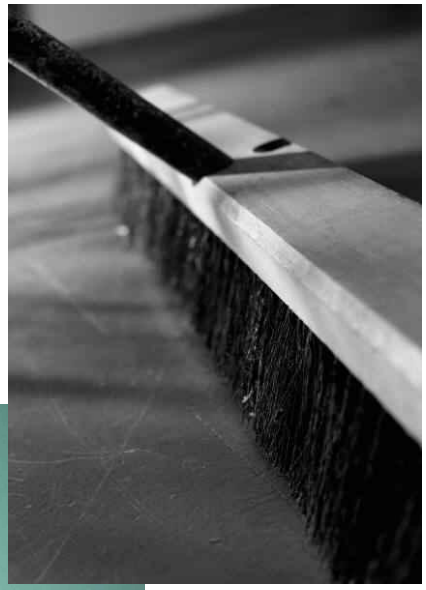
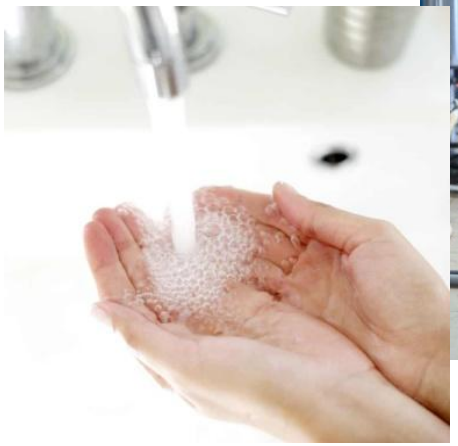
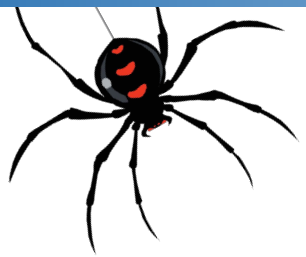
Maintain Facility

We are **<u>all responsible</u>** to help maintain the cleanliness of the facility, keeping it in Top Notch condition







Maintain your Facility
Pest Control

Pests: Includes but is not limited to rodents, insects, birds, and other organisms that may contaminate products or production areas. (See 4.13 Pest Control SOP)

Pest Control Measures: Actions taken to <u>prevent</u>, <u>monitor</u>, and eliminate pest infestations.

Elimination is done by licensed pest control contractor, currently use Clark Pest control.

Maintain the Facility
Pest Control

Preventative measures:

1. Sanitation

- Maintain cleanliness in all areas.
- Dispose of waste promptly and properly.
- Keep food and beverages out of production and storage areas.
- Keep the break room clean, store food in closed containers.

2. Structural Maintenance

- Seal cracks, gaps, and other potential entry points.
- Ensure doors and windows are fitted with proper seals and screens.

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Maintain the Facility
Pest Control

Preventative measures:

3. Storage Practices

- Store raw materials and finished products off the floor and away from walls.
- Use heat treated pallets.

4. Monitoring and Inspection

- Conduct weekly inspections for signs of pest activity (e.g., droppings, nests, damage).
- Report any signs of pests to your supervisor.

Maintain your Facility
Pest Control

- Be aware of pest control device locations
- Do not disturb pest control devices
- Keep outside doors closed
- 8 to 10” gap along exterior wall is required
- Maintain door seals-no light showing
- Report pest infestations to your supervisor
- Report poor seals, door closure issues, etc. to supervisor

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Maintain your Equipment - Preventive Maintenance

What is Preventative Maintenance?

- Maintenance performed regularly, before equipment malfunction and breakage.
- See 5.6.2 Equipment Maintenance SOP.
  Schedule and carry out the maintenance using the SOP guidelines.

Why is it important?

- To prevent equipment failure.

Preventive maintenance has many benefits:

- Enhance efficiency – well maintained instruments function better
- Extends equipment lifespan – saves operational cost
- Improve quality – equipment operates within specifications ensure optimal output.
- Reduce equipment downtime
- It save cost - helps to avoid costly repair.
- Improve safety – maintenance helps identify and eliminate potential safety hazards.

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Maintain your Equipment - Preventive Maintenance

Can we skip Preventive Maintenance?

No! It must be done regularly.

Examples of Preventative Maintenance.

- Perform routine cleaning and sanitizing of equipment.
- Check and replace worn or damaged parts.
- Perform routine lubrication of piston fillers before use by applying vitamin E.
- Routine lubrication and greasing of equipment.

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Maintain your Equipment Calibration

- Ensure measuring equipment is accurate and precise
- Must be documented
- Calibration standards must be traceable to a nationally recognized standard

- How do you know if the equipment is calibrated?

  Check the calibration records.

  Perform calibration check with know reference standard.

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Verify/Check
Incoming Inspection

Process for the inspection and verification of incoming

- raw materials,
- packaging materials,
- other goods
- Their documentation COA

The purpose is to ensure they meet specified requirements and are suitable for use in the production

Do not let damaged, contaminated goods enter our facility.

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Verify/Check Inspections

- Raw Materials
- Documentation: e.g. Certificate of Analysis COA
- Packing Materials
- Components
- Start-up/Make Ready
- In-process Inspections
- Final Inspections
- Batch Record Review
- Internal audit

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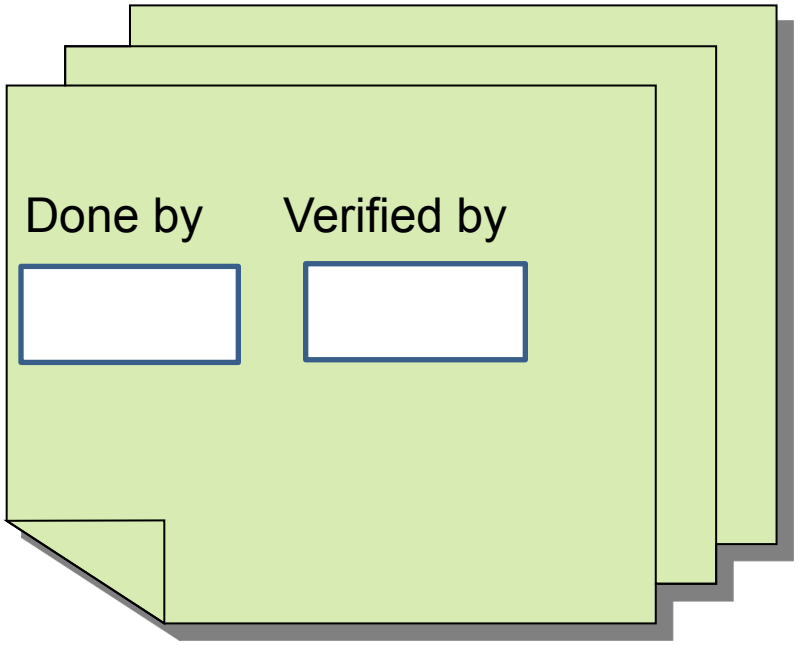
Verify/Check
Independent Second Party Verification



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Verify/Check
Line Clearance

Critical to preventing mixed product reaching our customers

Key Points:

- One job at a time in the workstation
- Staged work remains outside workstation
- Workstation defined by blue lines – includes workbench
- Line clearance performed and verified between jobs

- Inspect workstation for product, material, core/case labels, or paperwork from previous job **and remove any found**
- Inspect Machine for product/labels attached and remove.
- Second verification where required

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Verify/Check
Non-conformances

Non-conformity: A deviation from specified requirements or standards.

Related document: Certificate of Analysis (CoA)
Provided by the supplier that certifies the quality and specifications of the materials.

Non-Conforming Product: A product that does not meet all specified requirements for quality, function, or safety.

- It deviate from established specifications or drawings.
- Exhibit defects in materials, workmanship, or functionality.
- Fail to meet customer requirements.

Rejects must be clearly identified and segregated from acceptable product/components

Trends should be identified so that effective corrective action can be taken, preventing them from recurring

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Competency Requirements

- Individual must have a combination of experience, training and education that enables them to perform their assigned functions
  - General Knowledge
  - Job specific Knowledge
  - Education, Trade School, Seminars, other courses
- Training must be documented
  - Training Records
  - Competencies
  - Certifications

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Audit your Processes

- Types of Audits:
  - Internal Systems & Process Audits
  - External Customer Audits
  - Final Product Audit
- Why audit?
  - Provides accountability & recognizes gaps/non-conformances
  - It allows to fix problems and improve quality and safety
- Who should audit?
  - Trained Individuals independent of the production process.

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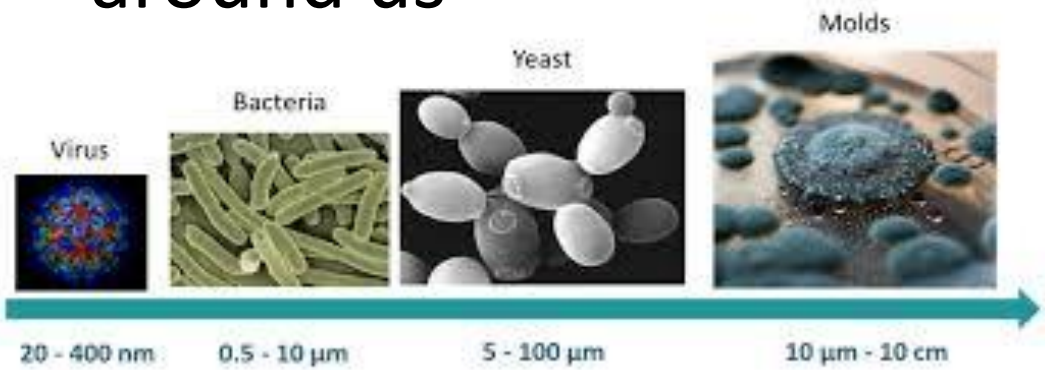
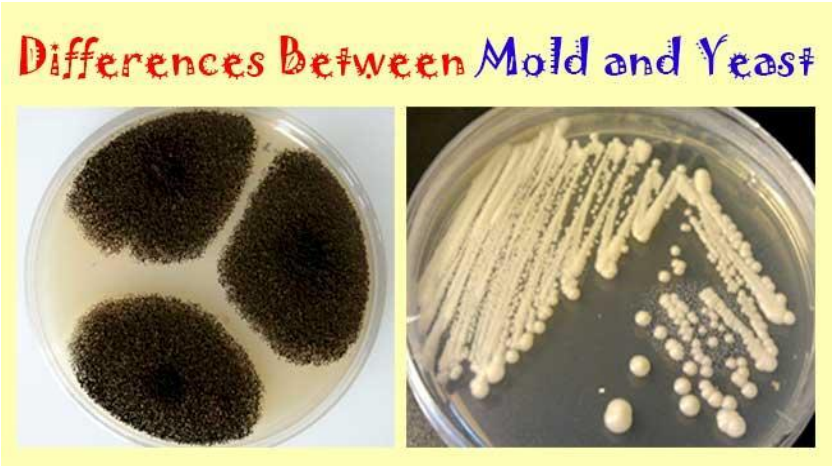
Microorganisms and Pathogens - on you and around us

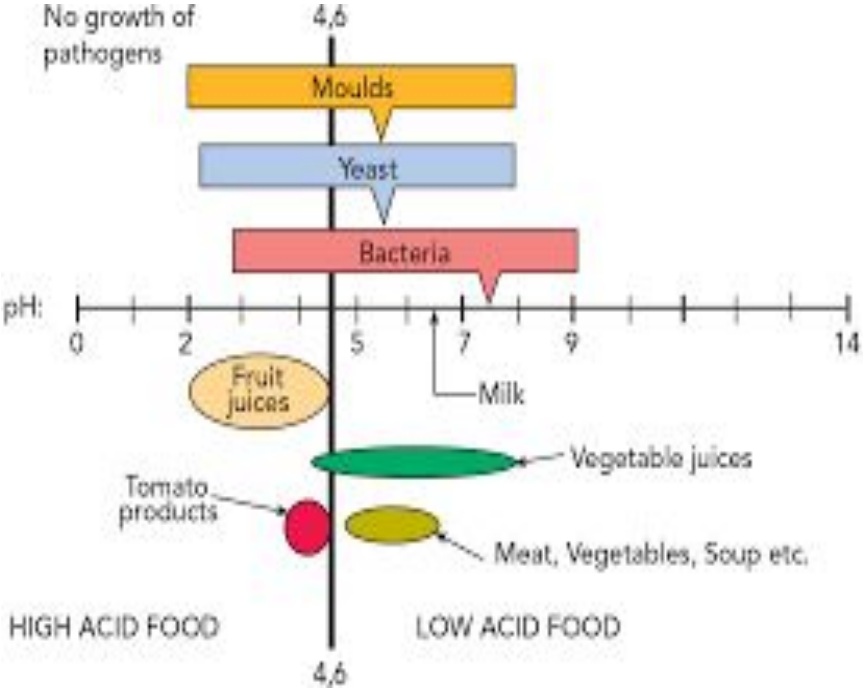




- Filamentous,
- Produce spore
- Can produce toxins

- Cluster of single cell
- No spores

Bacteria and molds grow in wide range of pH and substrate.



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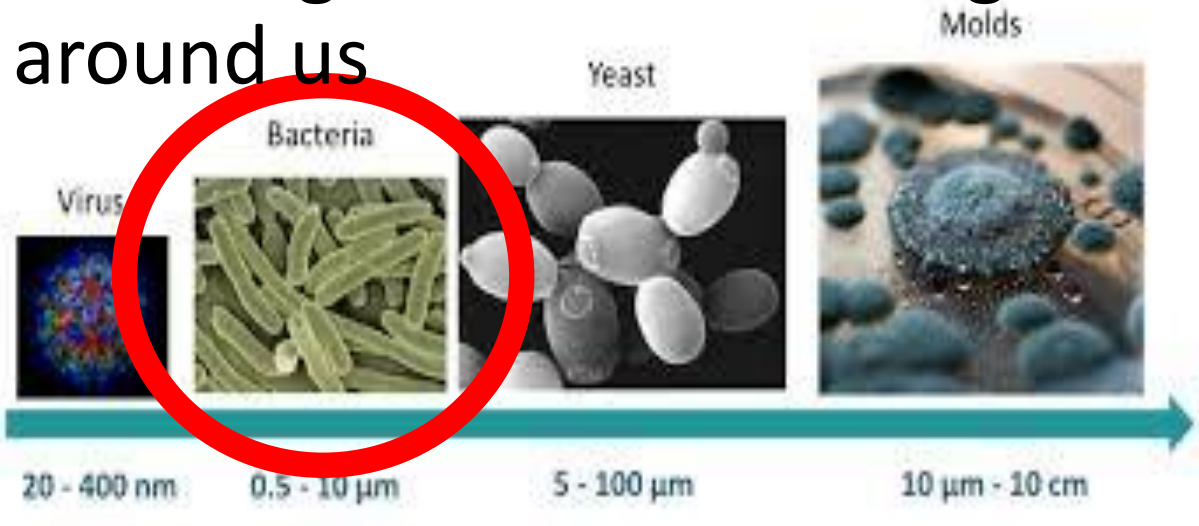
Microorganisms and Pathogens on you and around us



Bacterial growth and contamination in the equipment is a major issue.

1. fixed microorganisms in the system: live in cracks, gaps, and crevices or abrasive marks, insufficiently smoothed areas of the material, in dead spaces and form biofilms on equipment components.
2. 90% of bacterial contamination comes from the water system.

Bacteria and molds around us

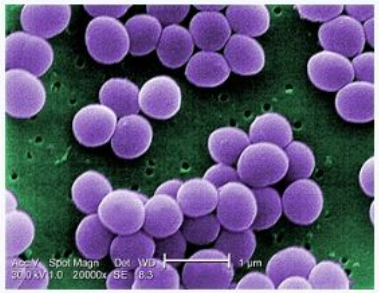
Pseudomonas aeruginosa: is a gram-negative, aerobic, non-spore forming rod shaped bacteria. Commonly found in the environment, particularly in **freshwater**. Reservoirs in urban communities include **hot tubs, jacuzzi, and swimming pools, and any stagnant water**. It can cause a wide array of community-acquired infections like folliculitis, puncture wounds leading to osteomyelitis, pneumonia, otitis externa, and many others. **It can invade our water system if not maintained well!**

Staphylococcus aureus is a gram-positive bacteria that cause a wide variety of diseases. It found in the environment and is also found in **normal human flora**, located on the **skin and mucous membranes** (most often the nasal area) of most healthy individuals.

***Escherichia coli* (*E. coli*)** is a gram-negative bacillus known to be a part of normal **intestinal flora** but can also be the cause of intestinal and extraintestinal illness. E. coli is mostly spread by dirty hands.

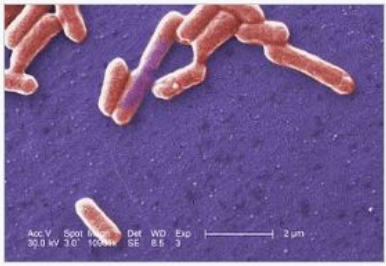
Pseudomonas aeruginosa on agar gel

Staphylococcus aureus on agar gel



Scanning electron micrograph of

***Escherichia coli* (*E. coli*) on agar gel**



Scanning electron micrograph of an *E. coli* colony

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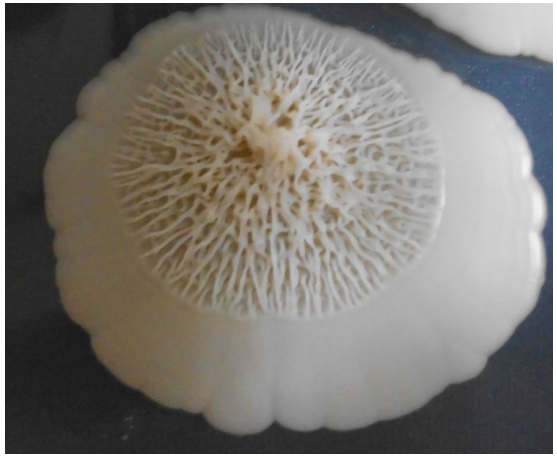
Bacteria and molds around us

Candida albicans is a common commensal fungus that colonizes the oropharyngeal cavity, gastrointestinal tract, and healthy **individuals' skin**. In 50% of the population, C. albicans is **part of the normal flora**.

Aspergillus brasiliensis is a fungus and is one of the most common species of the genus Aspergillus. It is ubiquitous **in soil**, a common **food contaminant**, and is also regularly reported from **indoor environment**s. As a fungal contaminant Aspergillus brasiliensis is transmitted through spores. The **spores** can be spread in manufacturing facilities via the hands of workers, or using equipment that isn't properly cleaned.

C. albicans growing as yeast-like cells with filamentous cells on top



C. albicans growing on agar gel

Aspergillus growing on agar gel

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Bacteria and molds - around us

Sanitation and cleanliness of the facility and equipment, and personal hygiene is vital to eliminate microorganisms from the production.

- Bacteria are carried on the **skin** as part of the normal skin flora.
- Bacteria are dispensed by **coughing** even if you are not sick.
- They are spread with **dirty hands**.
- Spores and bacteria can be carried with dirty, muddy, dusty **shoes** and **street cloth**.
- **Food**, lunchboxes, plates can carry mold and bacteria.
- **Cell phones** and other **personal items** such as watches, and jewelry are also covered with them.

<u>You must protect the production from germs!</u>

Wear the required attire and PPE!

Wash your hands!

Eat only in the designated areas!

Do not bring personal items to the production area!

Follow the cleaning and sanitizing procedures!

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Cleanliness
Personal Hygiene

- Keep hands clean when handling product and raw materials.
- Fingernails must be kept short and clean.
- Wear gloves as per regulations.
- Wash hands after eating, performing maintenance or using the restroom.
- Bandage cuts and open sores.
- Notify supervisor or manager if cuts or open sores occur while working with product or material.
- Notify supervisor if you are sick.

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Required dress code for the facility

Every time you enter or
leave the facility a new hairnet is issued

Production Area

Wear harirnet, gloves, shoe covers and coveralls <u>all the time</u> in the production area.

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Wear mask when you are filling a product

You must wear a mask when you are filling a product to protect the product from germs and bacteria.

The exhaled air contains tiny water droplets what can carry bacteria. These droplets can fly in the air for few meters. By wearing the mask you can stop the spread of these droplets. Mask will also protect the products if you need to sneeze.

The mask must cover your mouth and nose.

Place the mask over your mouth, nose, and chin, and tie it behind your head or use the ear loops.

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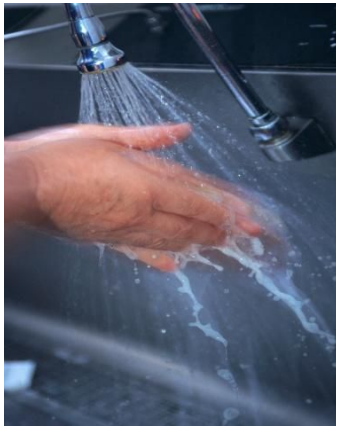
Cleanliness

No Food, Drink and Tobacco allowed in the production area.

This is required to prevent:

- Contamination of documents
- Contamination of raw materials
- Contamination of product
- Pest infestation



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Drinking is NOT allowed in the production area.

- Drinking is only allowed at the designated drink station.
- You are allowed to have only water at the drink station.

Water containers:

- No Glass, Ceramic/breakable containers allowed
- Screw-top plastic or metal container - 20 ounces or less with <1” opening
  - Remove empties at end of shift
  - Recycle or dispose in cafeteria / break area
- Keep in designated drink stations

Water must be consumed at designated drink stations

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Production Areas - Food

- Store in closed containers in cafeteria or locker

  Outside work area and away from product, documents and materials.

- Transport in closed containers through main aisles only.

  (coolers, lunch boxes, lunch bags – no glass or ceramic containers)

- Eat only in cafeteria or break area.

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Production Areas

Gum, medication and hard candy are **NOT** allowed in designated production areas.

- Notify your supervisor if you are required to take medication(s) during your shift.
- To ensure there are no safety issues/concerns, take necessary medications outside of designated production areas.
- Store medications in lockers.

- **Personal items**, such as jewelry and watches are **NOT** allowed in the production area.

They must be removed before entering production areas.

Cell phones and other devices

are NOT allowed in the production area.

Cell phones are usually dirty and carry a lot of bacteria.

- Avoid strong perfumes or colognes that could contaminate products.

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All Areas

- **Tobacco**
  - Store and transport in closed container
  - Smoke/chew only in designated outdoor areas
  - Keep smoking areas clean
  - Dispose of chewing waste in closed container outside
  - Wash hands after smoking

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Thank you for your attention!

Please ask questions.

References:

- Our Babylon Standard Operating Procedures
- Wilfried Rähse Cosmetic Creams Development, Manufacture and Marketing of Effective Skin Care Products 2020
- ISO 22716
- cGMP guidelines
- https://en.wikipedia.org/wiki/Pseudomonas_aeruginosa#/media/File:Pseudomonas_Aeruginosa.jpg
- https://en.wikipedia.org/wiki/Staphylococcus_aureus
- https://en.wikipedia.org/wiki/Escherichia_coli
- https://en.wikipedia.org/wiki/Candida_albicans